ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE B · MARKING GUIDELINE

MARKS: 80

Marking guideline for the Academic Central NC(V) Level 2 Electronics and Digital Concepts for Robotics paper 2 · practice b.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, equivalent designs and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 A ✓
1.2 C ✓
1.3 B ✓
1.4 B ✓
1.5 A ✓
1.6 D ✓
1.7 C ✓
1.8 A ✓
1.9 B ✓
1.10 D ✓

(10 × 1) [10]

QUESTION 2

2.1 C ✓
2.2 J ✓
2.3 I ✓
2.4 F ✓
2.5 E ✓
2.6 G ✓
2.7 H ✓
2.8 D ✓
2.9 B ✓
2.10 A ✓

(10 × 1) [10]

TOTAL SECTION A: [20]

SECTION B

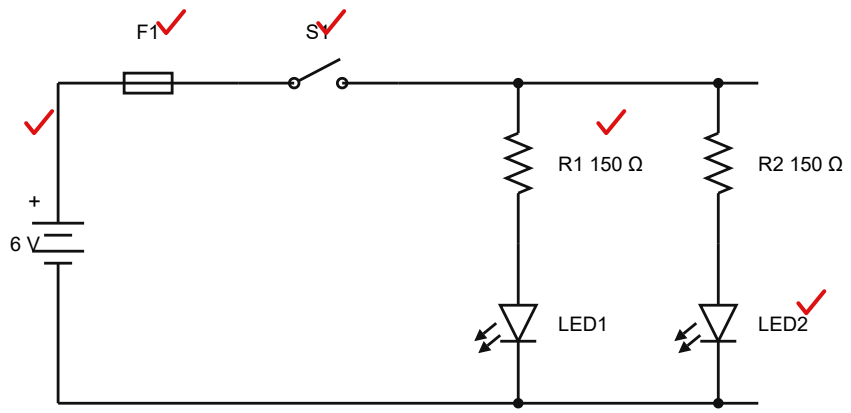
QUESTION 3

3.1.1 $R = (V_S - V_{LED}) \div I$ ✓ $= (6 - 3{,}2) \div 0{,}02 = 140\ \Omega$ ✓; choose $150\ \Omega$ ✓ (3)

3.1.2 $I_T = 20 + 20 = 40\ mA$ ✓ (1)

3.1.3 $P = I^2R \checkmark = 0{,}02^2 \times 140 = 0{,}06\ W \checkmark$ (2)

3.2



(Battery pack 1, fuse and switch in series with the supply 1, two parallel branches 1 each with a resistor in series with an LED 1, LEDs the right way round, neat with labels 1)

(5)

3.3 $V = 4 \times 1{,}5 = 6\ V \checkmark$; capacity stays 2 000 mAh ✓; $t = 2\ 000 \div 40 \checkmark = 50\ h \checkmark$ (4)

3.4 LEDs are never exactly the same; with one shared resistor the LED with the lowest voltage would take most of the current ✓, so it could burn out and the other one would be dim ✓. (2)

3.5 About 100 mA (a little above the normal 40 mA) ✓ so that it does not blow in normal use but blows on a short circuit. (Accept 63–250 mA with a reason.) (1)

3.6 Any TWO ✓✓: the longer leg is the anode; the flat side of the rim is the cathode; the larger plate inside is usually the cathode; test with the multimeter diode setting (it lights / conducts when red is on the anode). (2)

[20]

QUESTION 4

4.1 Connect the battery + and − to the long power rails ✓; place each resistor and LED so that they share a short row (the five holes in a row are connected) ✓; link each branch to the + and − rails with jumper wires ✓; keep the LED legs in different rows so they are not shorted, and check polarity ✓. (4)

4.2 In order ✓✓✓✓✓: clean the board and component leads; insert the component and bend the leads slightly; heat the pad and the lead together with a clean, tinned iron; feed solder onto the joint (not the iron) until it flows; remove the solder, then the iron, and let it cool without moving; trim the leads. (5)

4.3 Good: shiny, smooth, cone / volcano-shaped and covering the pad ✓. Dry: dull, grainy or a ball that does not wet the pad, may have cracks ✓. (2)

4.4 Setting ✓ and connection ✓ for each: (a) DC volts on a range above 6 V, probes across (in parallel with) the battery; (b) DC mA range, break the circuit and connect the meter in series with the LED; (c) ohms range, remove the resistor (or switch off the circuit) and put the probes across it. (6)

4.5 Any THREE ✓✓✓: LED in the wrong way round; dry joint or broken track; wrong resistor value (too high); faulty / blown LED; loose wire in the breadboard; flat battery or blown fuse (if both LEDs are off). (3)

[20]

QUESTION 5

5.1.1 $Y = T + A \cdot B$ ✓✓ (2)

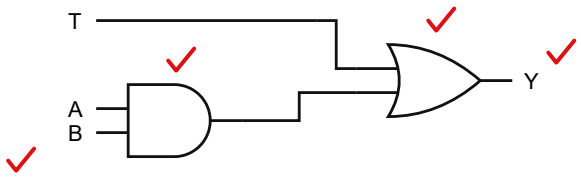
5.1.2

T	A	B	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

(8 rows 1; all T = 1 rows are 1 1; row 011 = 1 1; other T = 0 rows are 0 1. Any consistent column order.)

(4)

5.1.3



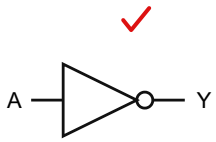
(AND gate with inputs A and B 1; OR gate with inputs T and the AND output 1; output labelled Y 1; correct symbols, neat 1)

(4)

5.2 (a) 0111 ✓✓ (b) 3 → 0011, 9 → 1001, so 0011 1001 ✓✓ (4)

5.3 Any TWO ✓✓: electronic switches have only two states (ON / OFF); two levels are easy to tell apart, so there are fewer errors from noise; simple, reliable circuits; easy to store and process. (2)

5.4



A	Y
0	1
1	0

(Triangle with a small circle at the output 1; truth table 0 → 1, 1 → 0 1)

(2)

5.5 Without it the input "floats" and can read random HIGH or LOW values from noise ✓; the pull-down resistor holds the input at 0 V (LOW) until the button connects it to the supply ✓. (2)

[20]

TOTAL SECTION B: [60]

TOTAL: 80

END OF MARKING GUIDELINE